Fair Offline Contextual Bandits with Guarantees under Inferred Attributes

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   Context: None
2. The submission template for submissions to RLJ/RLC 2025
   Context: Built from previous RLC/RLJ, ICLR, and TMLR submission templates
3. *[Example of one contribution and corresponding contextual note for the paper "Policy gradient methods for reinforcement learning with function approximation" (**?**).]*
   This paper presents an expression for the policy gradient when using function approximation to represent the action-value function.
   Context: Prior work established expressions for the policy gradient without function approximation (**?**).

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Abstract

Although offline contextual bandits achieve strong performance in low-stakes settings, their role in automated decision-making and decision support raises serious concerns about the fairness of their behavior. Many existing strategies for mitigating unfairness rely on regularization or empirical validation, which can fail to prevent spurious constraint satisfaction under finite data. One emerging strategy instead imposes probabilistic group-fairness constraints that explicitly account for statistical uncertainty in the data-generating process and its interaction with the learning pipeline. Existing high-confidence methods, however, assume that sensitive attributes (e.g., gender or age bracket) are observed as ground truth. In practice, these attributes are often unavailable and must be inferred. We show that naively treating inferred attributes as ground truth can invalidate high-probability fairness guarantees. We then develop methods for fair offline contextual bandits that restore valid guarantees under inferred attributes by incorporating assumptions on the reliability of the sensitive-attribute classifier. We focus on two common group fairness notions: demographic parity and reward disparity. Empirical evaluations on benchmark datasets demonstrate that our approach achieves competitive reward while satisfying the fairness guarantees.

1 Introduction

Modern machine learning systems are increasingly deployed in high-stakes sequential decision-making domains such as hiring, healthcare, and finance. In these settings, contextual bandit algorithms are commonly used to learn decision policies from logged interaction data. To mitigate harms, practitioners often impose fairness-regularized objectives or explicit group fairness constraints. However, enforcing such constraints introduces an issue: the learned policy is evaluated on the same data used to select it, creating a coupling between the learning procedure and the data-generating process. While conventional pipelines may tolerate modest overfitting to training data, this dependence can be problematic when fairness guarantees are required.

In practice, the system must often learn a candidate policy and verify its adherence to fairness constraints using the same pool of limited interaction data. Because optimization procedures exploit statistical fluctuations in finite samples, this reuse of data can produce the spurious satisfaction of fairness constraints. Such effects can lead to *fairness washing*, where discriminatory systems are falsely certified as compliant due to chance satisfaction of constraints during optimization (Aïvodji et al., 2019; Cotter et al., 2019). The problem is further exacerbated in complex pipelines involving multiple group-based checks, where the multiple comparisons problem increases the probability of certifying an unfair system as compliant as system complexity grows (Rupert Jr et al., 2012).

One framework for mitigating this issue enforces probabilistic constraints on system behavior that account for statistical uncertainty arising from the data-generating process and its interaction with the learning pipeline (Thomas et al., 2019; Chittepu et al., 2025). Crucially, this formulation typically assumes access to observed sensitive attributes (e.g., race, gender, or age), which are required to

compute group-based fairness constraints. In idealized settings, these attributes are verified and exhaustive, providing reliable ground truth. However, in large-scale deployments, sensitive attributes are often unavailable. For example, social media platforms may intentionally avoid collecting such information during user registration due to privacy regulations, ethical considerations, or to reduce onboarding friction [add citation(s)]. The absence of explicit sensitive attribute labels fundamentally limits the applicability of fairness-aware learning methods, which typically rely on this information during training or evaluation. To circumvent this limitation, missing sensitive attributes are often imputed using heuristic methods [add citation(s)], or predicted via auxiliary classifiers trained on correlated features such as names, geographic locations, or behavioral patterns [add citation(s)].

While these *inferred labels* allow for the computation of standard fairness constraints, they fundamentally alter the statistical setting described above. Probabilistic guarantees rely on accurate estimation of group-dependent quantities; when group membership itself is uncertain, these high-confidence guarantees may no longer hold with respect to the true underlying attributes. As a result, the algorithm may incorrectly certify a policy as satisfying the desired fairness constraints, or behave overly conservatively due to inflated uncertainty. In either case, using inferred attributes can compromise the validity and calibration of fairness guarantees in offline contextual bandits.

In this work, we develop algorithms for offline contextual bandits with high-probability fairness guarantees. First, we present an algorithm that maximizes expected reward subject to three fairness notions: demographic parity, reward disparity, and minimum group reward constraints. Second, we study the setting in which some sensitive attributes are inferred rather than observed. We show that naively treating inferred attributes as ground truth can invalidate high-probability fairness verification, leading to false guarantees or unnecessary conservatism. To address this issue, we propose methods that restore valid fairness guarantees with respect to the latent attributes under appropriate assumptions. Empirically, we demonstrate that our methods preserve high-confidence guarantees while achieving competitive reward compared to baseline methods.

2 Problem Setting

We consider the offline contextual bandit setting under group fairness constraints. Let $\mathcal{X}$ denote the context space and $\mathcal{A}$ a finite action space. Each instance is characterized by a context $X \in \mathcal{X}$ and a sensitive attribute $S \in \mathcal{S} = \{s_1, s_2\}$ representing group membership. Context–attribute pairs (X, S) are drawn from an unknown joint distribution $P_{X,S}$. Given a context X, an action $A \in \mathcal{A}$ is selected according to a behavior policy $\pi_b(\cdot \mid X)$, and a reward $R \in [0, R_{\max}]$ is generated according to an unknown conditional distribution $P_{R|X,S,A}$.

Let $\{\pi_\theta : \theta \in \Theta\}$ denote a parameterized class of target policies, where each $\pi_\theta : \mathcal{X} \to \Delta(\mathcal{A})$. For any $\theta \in \Theta$, we define the expected reward of π_θ as $r(\theta) := \mathbb{E}[R]$, where the expectation is taken over $(X, S) \sim P_{X,S}$, $A \sim \pi_\theta(\cdot \mid X)$, and $R \sim P_{R|X,S,A}$. To capture group-level outcomes, we define the group-conditional expected reward for each $s \in \mathcal{S}$ as $r_s(\theta) := \mathbb{E}[R \mid S = s]$. We consider two common group-based notions of fairness that arise in contextual decision-making settings. These capture disparities in outcomes and disparities in action allocation.

Definition 1 (Group-based fairness constraints)**.** *Let θ parameterize a policy π_θ, and let $r_s(\theta)$ denote the expected reward conditional on group $S = s$.*

1. **Reward disparity:** *For tolerance $\epsilon > 0$, reward disparity requires $|r_{s_1}(\theta) - r_{s_2}(\theta)| \leq \epsilon$.*
2. **Demographic parity:** *Let $a^+ \in \mathcal{A}$ denote a designated beneficial action and define the group-conditional selection probability $p_s(\theta) := \mathbb{E}[\mathbb{I}\{A = a^+\} \mid S = s]$, where $(X, S) \sim P_{X,S}$ and $A \sim \pi_\theta(\cdot \mid X)$. Demographic parity requires $|p_{s_1}(\theta) - p_{s_2}(\theta)| \leq \epsilon$.*

2.1 High-Probability Fairness-Constrained Learning

In the offline setting, we observe an i.i.d. logged dataset $D = \{(X_i, S_i, A_i, R_i)\}_{i=1}^n$, where $(X_i, S_i) \sim P_{X,S}$, $A_i \sim \pi_b(\cdot \mid X_i)$, and $R_i \sim P_{R|X_i,S_i,A_i}$. Our goal is to learn a policy that

maximizes expected reward while satisfying one of the fairness criteria defined in Definition 1. Let $g(\theta)$ denote a population-level constraint function corresponding to one of these fairness criteria (e.g., reward disparity, demographic parity, or minimum group performance), calibrated such that $g(\theta) \leq 0$ implies the fairness requirement under the true data-generating distribution.

In practice, the quantities defining g depend on the unknown data-generating distribution and must therefore be estimated from the finite dataset D. Consequently, learning algorithms must account for the statistical uncertainty arising from finite logged data when selecting a policy that satisfies the desired fairness constraint. To address this challenge, we consider algorithms that enforce *probabilistic fairness guarantees* (Thomas et al., 2019). Under this requirement, the returned policy must satisfy the chosen fairness constraint under the true data-generating distribution with high probability over the randomness of the offline dataset. Formally, let $\theta(D)$ denote the parameter returned by a learning algorithm given dataset D. For a user-specified failure probability $\delta \in (0, 1)$, we seek algorithms satisfying

$$\Pr_{D}\big(g(\theta(D)) \leq 0\big) \geq 1 - \delta, \tag{1}$$

where the probability is taken with respect to the data-generating process $(P_{X,S}, \pi_b, P_{R|X,S,A})$ that governs the logged dataset. We refer to this requirement as the *probabilistic fairness guarantee*.

Algorithm 1

Require: Dataset D; Confidence level δ
Ensure: Candidate θ_c or NSF
1: $D_{\text{CS}}, D_V \leftarrow \text{Partition}(D)$
2: $\theta_c \leftarrow \text{CandidateSelection}(\delta, D_{\text{CS}})$
3: $\hat{g} = \phi(D_V)$ ▷ Verification Step
4: **if** $U(\hat{g}) > 0$ **return** NSF **end if**
5: **return** θ_c

Algorithms satisfying the probabilistic fairness guarantee typically follow a three-step protocol, summarized in Algorithm 1 [add citation(s)]. First, the **partition step** (line 1) splits the training dataset D into a candidate selection set D_{CS} and a verification set D_V. Next, the **candidate selection step** (line 2) uses D_{CS} to optimize a candidate solution θ_c. Finally, in the **verification step** (lines 3–7), the algorithm uses D_V to estimate each constraint value via $\hat{g} = \phi(D_V)$ and computes a high-confidence upper bound (HCUB), denoted $U(\cdot)$. If any bound satisfies $U(\hat{g}) > 0$, the candidate is rejected and the algorithm returns "No Solution Found" (NSF); otherwise θ_c is returned. Allowing the algorithm to abstain via NSF is critical for fairness; it prevents the system from being forced to deploy a policy in which it lacks statistical confidence, such as when data is insufficient or constraints are conflicting (e.g., in group fairness settings (**??**)[add citation(s)]). We treat abstention (NSF) as satisfying all fairness constraints by convention, i.e., $g(\text{NSF}) = 0$.

The candidate selection step performs a domain-specific optimization on D_{CS} to identify a high-utility candidate solution. While the specific optimization depends on the application domain, in contextual bandit settings this typically corresponds to approximately maximizing expected reward [add citation(s)]. This step is primarily responsible for performance and may vary across problem settings, as it can use any suitable optimization or heuristic procedure to produce a promising candidate. Importantly, candidate selection need not enforce the fairness guarantee directly; as we show in Section 3, this responsibility is handled by the subsequent verification step. As a result, many algorithms that satisfy (1) share a common verification procedure, which we describe next.

The Verification Step. The verification step uses the held-out dataset D_V to construct a high-confidence upper bound on the population constraint value $g(\theta_c)$, where θ_c is the model returned by the candidate selection procedure. For the group-based fairness constraints in Definition 1, verification proceeds by first estimating a signed group-difference quantity and then converting this estimate into a bound on the constraint value. Specifically, define the signed disparity. Both fairness notions can be written in the unified form $g(\theta) = |\Delta(\theta)| - \epsilon$.

For θ_c, the verifier computes i.i.d. per-sample statistics $Z_1(\theta_c), \ldots, Z_m(\theta_c)$ from the rows of D_V such that $\mathbb{E}[Z_i(\theta_c) \mid \theta_c] = \Delta(\theta_c)$. The verification statistic in Algorithm 1 is

$$\hat{g} = \phi(D_V) = \frac{1}{m} \sum_{i=1}^{m} Z_i(\theta_c),$$

where the dependence of ϕ on the fixed candidate θ_c is suppressed for notational simplicity. Thus, $\hat{g}$ is an unbiased estimator of $\Delta(\theta_c)$. A confidence interval is then constructed for $\Delta(\theta_c)$ using $\hat{g}$ and the empirical variance of the $Z_i(\theta_c)$, and this interval is mapped to a bound on $g(\theta_c)$ via the relation $g(\theta) = |\Delta(\theta)| - \epsilon$.

Different statistical tools can be used to construct such bounds. In this section we present an example based on Student's t-test (Student, 1908), and defer a construction based on Hoeffding's inequality (Hoeffding, 1963) to Appendix B.

Consider a vector of m independent and identically distributed (i.i.d.) samples $(Y_i)_{i=1}^m$ of a random variable Y. Let the sample mean be $\bar{Y} = \frac{1}{m}\sum_{i=1}^m Y_i$, and the sample standard deviation with Bessel's correction be $\sigma(Y_1, \ldots, Y_m) = \sqrt{\frac{1}{m-1}\sum_{i=1}^m (Y_i - \bar{Y})^2}$. Let $\delta \in (0, 1)$ denote a confidence level.

Property 1 (Student's t-test)**.** *Let $t_{1-\delta,m-1}$ be the $1-\delta$ quantile of the Student's t distribution with $m-1$ degrees of freedom. If $\bar{Y}$ is normally distributed, then*

$$\Pr\left(\mathbb{E}[Y] \leq \bar{Y} + \frac{1}{\sqrt{m}}\sigma(Y_1, \ldots, Y_m)\, t_{1-\delta,m-1}\right) \geq 1 - \delta.$$

Proof. See Student (1908). □

Property 1 yields a high-confidence upper bound on the mean of Y:

$$U_{\text{ttest}}(Y_1, \ldots, Y_m) \coloneqq \bar{Y} + \frac{1}{\sqrt{m}}\sigma(Y_1, \ldots, Y_m) t_{1-\delta,m-1}.$$

Applying this result with $Y_i = Z_i(\theta_c)$ gives

$$\Pr\big(\Delta(\theta_c) \leq U_{\text{ttest}}(Z_1(\theta_c), \ldots, Z_m(\theta_c))\big) \geq 1 - \delta.$$

Since $g(\theta_c) = |\Delta(\theta_c)| - \epsilon$, this induces a valid high-confidence upper bound on $g(\theta_c)$ via the same transformation.

Thus, the HCUB computed in line 4 of Algorithm 1 can be instantiated using U_{ttest} when the corresponding assumptions are satisfied. Importantly, the validity of a particular confidence-bound construction depends on the assumptions under which it is derived. For instance, Student's t-test yields exact bounds when the sample mean is normally distributed. By the Central Limit Theorem, this assumption is approximately satisfied for sufficiently large m, as the sample mean converges to a normal distribution regardless of the distribution of Y_i.

2.2 Inferred Attribute Regime

In many practical deployments, sensitive attributes are not fully observed. We therefore consider a setting in which the available dataset consists of two subsets, $D = D_{\text{gt}} \cup D_{\text{inf}}$, where $D_{\text{gt}} = \{(X_i, S_i, A_i, R_i)\}_{i=1}^{n_{\text{gt}}}$ contains observed sensitive attributes, and $D_{\text{inf}} = \{(X_j, \widehat{S}_j, A_j, R_j)\}_{j=1}^{n_{\text{inf}}}$ contains inferred attributes in place of the ground-truth labels.

The inferred attribute $\widehat{S}$ is produced by a fixed prediction rule $\widehat{S} \sim h(\cdot \mid X)$, where $h : \mathcal{X} \to \Delta(\mathcal{S})$ maps contexts to a distribution over sensitive attributes. Because the fairness constraints are defined with respect to the true latent attribute S, we require structural and statistically verifiable conditions on the attribute predictor h. In particular, we assume that the predictor used to generate inferred attributes satisfies at least one of the following properties.

Assumption 1. *The prediction rule h is trained using data independent of D_{gt} and D_{inf}. Conditional on h, the samples in D_{gt} and D_{inf} are independent draws from the underlying data-generating process.*

Assumption 2. *The predictor h satisfies* at least one *of the following conditions:*

1. ***Bounded confusion matrix:*** *Let $C_{s,s'} := \Pr(\widehat{S} = s' \mid S = s)$. The confusion matrix C lies in a known uncertainty set $\mathcal{C}$ defined by entrywise bounds together with row-stochasticity constraints.*
2. ***Per-group bounded misclassification error:*** *There exist constants $\varepsilon_s \in [0, 1)$ such that*

$$\Pr(\widehat{S} \neq S \mid S = s) \leq \varepsilon_s, \qquad \forall s \in \mathcal{S}. \tag{2}$$

These conditions reflect standard auditing practices for attribute predictors. Group-conditional misclassification rates and confusion matrices are routinely estimated on independent labeled datasets and reported with confidence intervals, making the corresponding bounds empirically verifiable. Because these quantities can be estimated with statistical guarantees, they can be incorporated into the verification procedure used by algorithms enforcing the probabilistic fairness guarantee (e.g., (Thomas et al., 2019)). As we show in Section 3, requiring that at least one of these properties hold allows the algorithm to account for uncertainty in inferred attributes while maintaining the probabilistic fairness guarantee.

3 Method and Theoretical Analysis

In Section 3.1, we present a baseline offline contextual bandit algorithm that satisfies the probabilistic fairness guarantee in (1) assuming only ground truth data D_{gt} is provided. We then analyze two practical scenarios that arise when sensitive attributes are inferred rather than directly observed. First, in Section 3.2, we consider the setting where inferred samples are used only in the candidate selection step. We show that this modification preserves the fairness guarantee while altering the optimization problem. Finally, in Section 3.3, we consider the setting where inferred attributes appear in both the candidate and verification sets. In this case both the candidate selection procedure and the verification step must be modified in order to maintain the fairness guarantee. All proofs are deferred to Appendix [TODO:].

3.1 High Confidence Fair Offline Contextual Bandit (HC-Fair)

Algorithm 2 HC-Fair

Require: Ground-truth dataset D_{gt}; confidence level δ
Ensure: Candidate θ_c or NSF
1: $D_{\text{CS}}, D_V \leftarrow \text{Partition}(D_{\text{gt}})$
2: $\theta_c \leftarrow$ [TODO: YASWANTH] ▷ Candidate Selection
3: **return** $\text{VerificationStep}(\theta_c, D_V, \delta)$

Algorithm 2 presents our baseline High-Confidence Fair offline contextual bandit algorithm (HC-Fair). The ground-truth dataset D_{gt} is first randomly partitioned into a candidate selection set D_{CS} and a verification set D_V. A candidate policy θ_c is learned using D_{CS}, [TODO: Describe candidate selection process.] after which the verification procedure from Section 2.1 is used to determine whether it is returned. We therefore focus below on the candidate selection method.

3.1.1 Inferred Attribute Candidate Selection

[Todo: clean –Blossom]

Suppose we have a dataset $D = D_{gt} \cup D_{in}$. D_{gt} is the subset of data for which we have access to the ground-truth sensitive attributes. D_{in} is the subset of data for which the sensitive attribute labels

are predicted by a classifier. Futhermore, these two subsets are disjoint i.e $D_{gt} \cap D_{in} = \emptyset$. We assume we know the identity of datapoints beforehand i.e we know which data is in D_{gt} and which data is in D_{in}. We define a variable Z that denotes the subset of data we are considering. Let's say $Z = 1$ implies that we are considering D_{gt} and $Z = 0$ implies that we are considering D_{in}. Furthermore, let S be the variable representing the sensitive attribute and $\hat{S}$ represents the variable representing the noisy predicted sensitive attribute. We point out that for D_{in}, S is unobserved and $\hat{S}$ is unknown.

Additionally, we make an assumption on the max possible error rate of a classifier used to infer the sensitive attributes

$$P(\hat{S} \neq S | Z = 0) \leq \epsilon \tag{3}$$

We further assume that we only have two sensitive attributes in the dataset A, B. Let X be the random variable of interest to us, for the fairness constraint. When looking at reward disparity between the sensitive attribute groups, X represents the reward. When looking at action disparity instead, X represents the action.

We are interested in fairness constraints of the form

$$\big|\mathbb{E}[X|S = A] - \mathbb{E}[X|S = B]\big| \leq \epsilon \tag{4}$$

First, we need to be able to compute $\mathbb{E}[X|S = A]$. We discuss how to compute this expectation now. By the law of total expectation, we have:

$$\mathbb{E}[X|S = A] = \mathbb{E}[X|S = A, Z = 1]P(Z = 1|S = A) + \mathbb{E}[X|S = A, Z = 0]P(Z = 0|S = A) \tag{5}$$

We assume that the identity of the sensitive attribute does not tells us anything about the type of the dataset i.e $Z \perp S$. Hence $P(Z = 1|S = A) = P(Z = 1)$. We set $\lambda = P(Z = 1)$. The expectation then becomes

$$\mathbb{E}[X|S = A] = \mathbb{E}[X|S = A, Z = 1]\lambda + \mathbb{E}[X|S = A, Z = 0](1 - \lambda) \tag{6}$$

$\mathbb{E}[X|S = A, Z = 1]$ can be easily computed from D_{gt}, since D_{gt} contains information about the sensitive attribute S. The problem is with computing $\mathbb{E}[X|S = A, Z = 0]$, because for D_{in} we only have access to the noisy predicted inferred attribute $\hat{S}$, while S is unobserved.

Since we only have access to $\hat{S}$, when $Z = 0$, consider the expectation $\mathbb{E}[X|\hat{S} = A, Z = 0]$. Applying the law of total expectation, conditioning on S, we have

$$\begin{aligned}\mathbb{E}[X|\hat{S} = A, Z = 0] &= \mathbb{E}[X|\hat{S} = A, S = A, Z = 0] \cdot P(S = A|\hat{S} = A, Z = 0) \\ &\quad + \mathbb{E}[X|\hat{S} = A, S = B, Z = 0] \cdot P(S = B|\hat{S} = A, Z = 0) \qquad (7) \\ asdfasd & \qquad (8)\end{aligned}$$

We make an important assumption that $X \perp \hat{S} \mid S, Z$ i.e X is conditionally independent of $\hat{S}$, given S, Z. Using this conditional independence assumption, we can express the expectation as

$$\begin{aligned}\mathbb{E}[X|\hat{S} = A, Z = 0] &= \mathbb{E}[X|S = A, Z = 0] \cdot P(S = A|\hat{S} = A) \\ &\quad + \mathbb{E}[X|S = B, Z = 0] \cdot P(S = B|\hat{S} = A) \qquad (9)\end{aligned}$$

Let $\mu_A = \mathbb{E}[X|S = A, Z = 0]$ and $\mu_B = \mathbb{E}[X|S = B, Z = 0]$. We can express the expectation as

$$\mathbb{E}[X|\hat{S} = A, Z = 0] = \mu_A \cdot P(S = A|\hat{S} = A) + \mu_B \cdot P(S = B|\hat{S} = A) \tag{10}$$

Estimating $P(S \mid \hat{S})$. We now compute $P(S = A \mid \hat{S} = A)$ and $P(S = B \mid \hat{S} = A)$. Let

$$\pi_A = P(S = A), \qquad \pi_B = P(S = B).$$

We assume that the split between D_{gt} and D_{in} is *stratified* with respect to the sensitive attribute. Consequently, the distribution of S is preserved across the subsets:

$$P(S = A) = P(S = A \mid Z = 1) = P(S = A \mid Z = 0) = \pi_A,$$
$$P(S = B) = P(S = B \mid Z = 1) = P(S = B \mid Z = 0) = \pi_B.$$

Under this assumption, the group proportions estimated from D_{gt} are representative of the entire dataset and can therefore be used as estimates of the true priors π_A and π_B.. From Bayes rule, we have

$$P(S = A \mid \hat{S} = A) = \frac{P(\hat{S} = A \mid S = A)P(S = A)}{P(\hat{S} = A)} = \frac{(1-\epsilon)\pi_A}{(1-\epsilon)\pi_A + \pi_B} \tag{11}$$

Similarly, we have

$$P(S = B \mid \hat{S} = A) = \frac{P(\hat{S} = A \mid S = B)P(S = B)}{P(\hat{S} = A)} = \frac{\epsilon\pi_B}{(1-\epsilon)\pi_A + \epsilon\pi_B} \tag{12}$$

$$P(S = A \mid \hat{S} = B) = \frac{P(\hat{S} = B \mid S = A)P(S = A)}{P(\hat{S} = B)} = \frac{\epsilon\pi_A}{\epsilon\pi_A + (1-\epsilon)\pi_B} \tag{13}$$

$$P(S = B \mid \hat{S} = B) = \frac{P(\hat{S} = B \mid S = B)P(S = B)}{P(\hat{S} = B)} = \frac{(1-\epsilon)\pi_B}{\epsilon\pi_A + (1-\epsilon)\pi_B} \tag{14}$$

Similar to Equation 10, we can come up with an analogous equation, conditioned on $\hat{S} = B$. This gives us the system of equations

$$\mathbb{E}[X|\hat{S} = A, Z = 0] = \mu_A \cdot P(S = A|\hat{S} = A) + \mu_B \cdot P(S = B|\hat{S} = A) \tag{15}$$
$$\mathbb{E}[X|\hat{S} = B, Z = 0] = \mu_A \cdot P(S = A|\hat{S} = B) + \mu_B \cdot P(S = B|\hat{S} = B) \tag{16}$$

$\mathbb{E}[X|\hat{S} = A, Z = 0]$ and $\mathbb{E}[X|\hat{S} = B, Z = 0]$ can be easily inferred from D_{in}. Let $\hat{\mu} = \Big[\mathbb{E}[X|\hat{S} = A, Z = 0], \mathbb{E}[X|\hat{S} = B, Z = 0]\Big]$. Let $\mu = \Big[\mathbb{E}[X|S = A, Z = 0], \mathbb{E}[X|S = B, Z = 0]\Big] = \Big[\mu_A, \mu_B\Big]$.

Let M be the matrix

$$M = \begin{bmatrix} P(S = A|\hat{S} = A) & P(S = B|\hat{S} = A) \\ P(S = A|\hat{S} = B) & P(S = B|\hat{S} = B) \end{bmatrix} = \begin{bmatrix} \frac{(1-\epsilon)\pi_A}{(1-\epsilon)\pi_A + \epsilon\pi_B} & \frac{\epsilon\pi_B}{(1-\epsilon)\pi_A + \epsilon\pi_B} \\ \frac{\epsilon\pi_A}{\epsilon\pi_A + (1-\epsilon)\pi_B} & \frac{(1-\epsilon)\pi_B}{\epsilon\pi_A + (1-\epsilon)\pi_B} \end{bmatrix}$$

Then we estimate $\mu = M^{-1}\hat{\mu}$. This allows us to compute the group conditioned expectation in Equation 6. Similarly we can compute the expectation conditioned on $S = B$. We now have estimated the values of $\mathbb{E}[X|S = A]$ and $\mathbb{E}[X|S = A]$

Overall, to compute the group-conditioned expectations, we split this expectation into three conditional expectations. Lets represent this as

$$\mathbb{E}[X|S = A] = p_1\mathbb{E}[X|S = A, Z = 1] + p_2\mathbb{E}[X|\hat{S} = A, Z = 0] + p_3\mathbb{E}[X|\hat{S} = B, Z = 0] \quad (17)$$

The constants p_1, p_2, p_3 are described in the derivation earlier.

Bound Propagation and Optimization We use the Bound Propagation technique in Robinhood (**?**), to compute the upperbound of our absolute difference of group conditioned expectations. Lets call this upper bound U. We first reiterate our optimization objective.

$$\max_{\theta} \mathbb{E}_{x\sim D, y\sim\pi_b(.|x)}\left[\frac{\pi_\theta(y|x)}{\pi_b(y|x)} r(x, y)\right] \quad (18)$$

$$s.t \quad \left|\mathbb{E}_{x\sim D, y\sim\pi_b(.|x)}\left[\frac{\pi_\theta(y|x)}{\pi_b(y|x)} r(x, y)|S = A\right] - \mathbb{E}_{x\sim D, y\sim\pi_b(.|x)}\left[\frac{\pi_\theta(y|x)}{\pi_b(y|x)} r(x, y)|S = B\right]\right| \leq \epsilon \quad (19)$$

We list out two optimization routines. One a black box approach, using CMA-ES. The other variant optimizes the Lagrangian using Dual Ascent. We briefly describe both here

CMA-ES We use bound propagation to calculate the upper bound of the fairness constrint. Lets call this U. Since, CMA minimzes by default, whenever $U > \epsilon$ we return a very high value (essentially enforcing the constraint as a hard barrier function), and when the constrained is satisfied we return the negative Expected IS return.

Lagrangian We split the absolute fairness constraint into two constraints, using the property of the absolute value function i.e $|x - y| \leq \epsilon \implies x - y \leq \epsilon \quad \wedge \quad y - x \leq \epsilon$. We the use bound propagation (**?**), to compute the upper-bounds of these two constraints. The gradient of the upper-bound wrt policy parameters can be easily computed using the REINFORCE trick. The Lagrangian of this problem is then optimized using Dual Ascent.
[END –Blossom]

We now show that HC-Fair satisfies the probabilistic fairness guarantee. To do so, we first show a more broad result that establishes the probabilistic fairness guarantee for methods adhering to Algorithm 1 reduces to proving the validity of the HCUB used in the verification step.

Theorem 1. *Consider Algorithm 1 with D_{gt} as input. Let θ_c be the candidate policy produced using D_{CS}, and let $U(\hat{g})$ be the high-confidence upper bound of $g(\theta_c)$ computed on D_V. If $\Pr\big(g(\theta_c) > U(\hat{g})\big) \leq \delta$, then Algorithm 1 satisfies the fairness guarantee in* (1).

This reduction isolates the guarantee to the verification step: if the high-confidence bound is valid, the overall fairness guarantee follows regardless of the candidate selection procedure (e.g., whether θ_c is derived via large-scale pre-training, RLHF, or prompting strategies). To apply Theorem 1 to our setting, we must characterize when the HCUB constructed in the verification step is statistically valid. For concreteness, we instantiate the verification step using Student's t-test (Property 1) to construct confidence intervals. This requires the following normality assumption.

Assumption 3 (Normality Assumption for t-test)**.** *The sample mean $\mathrm{Avg}(\hat{g}) = \frac{1}{m}\sum_{i=1}^{m}\hat{g}_i$ is normally distributed.*

We additionally assume that the dataset partition used in line 1 of Algorithm 2 is random.

Assumption 4 (Random Partitioning)**.** *Algorithm 2 partitions the dataset D into disjoint subsets D_{CS} and D_V using a random split independent of the data values.*

Theorem 2. *Let g be defined as in Definition 1. Under Assumptions 3 and 4,* $\Pr(g(\mathrm{alg}(D)) \leq 0) \geq 1 - \delta$*, where* alg *denotes Algorithm 2.*

Other confidence interval constructions may also be used in place of the t-test. For example, if Hoeffding's inequality (Property 2 in Appendix B) were used, the corresponding assumption would instead require the estimates $\hat{g}_i$ to be bounded.

3.2 Inferred Attributes Only in Candidate Set

Many algorithms that enforce the probabilistic fairness guarantee assume that the sensitive attributes in the training data are observed ground truth [add citation(s)]. In practice, however, modern ML pipelines often operate in settings where sensitive attributes are partially missing. In such cases, a smaller ground-truth dataset is frequently supplemented with additional samples whose attributes are inferred, such as model-predicted demographic labels used for fairness auditing or AI-generated feedback used to augment training data (e.g., RLAIF) [add citation(s)].

A naive application of Algorithm 2 to datasets containing inferred attributes can invalidate the verification guarantees. In particular, applying the random partitioning strategy from Assumption 4 to the full dataset allows samples with inferred attributes to enter the verification set D_V. As a result, the verification data are drawn from a mixture of ground-truth and inferred-attribute distributions rather than a single data-generating process. Since standard HCUB constructions assume identically distributed verification samples (**?**)[add citation(s)], the resulting verification estimates $\{\hat{g}_i\}$ may violate the assumptions required for valid coverage. We formalize this failure mode below.

Lemma 1 (Failure of Naive Verification with Inferred Attributes)**.** *Let $D = D_{gt} \cup D_{inf}$, where $D_{inf} \neq \emptyset$. If Algorithm 2 is applied under Assumption 4, then* (1) *need not hold.*

This failure arises from allowing samples with inferred attributes to enter the verification set, thereby violating the distributional assumptions required by standard HCUB methods (**?**) [add citation(s)]. One way to restore validity is therefore to restrict verification to ground-truth data only. Specifically, we construct the verification set using only ground-truth samples ($D_V \subseteq D_{\text{gt}}$), while allowing the candidate selection set to leverage all remaining data:

$$D_{\text{CS}} = (D_{\text{gt}} \setminus D_V) \cup D_{\text{inf}}. \tag{20}$$

This modification allows the candidate selection step to benefit from additional samples with inferred attributes, while ensuring that the verification step operates only on ground-truth data. We show in Theorem [TODO: **??**] that this preserves the probabilistic fairness guarantee.

Algorithm 3 HC-Fair (Inferred Attributes in Candidate Set)

Require: Dataset $D = D_{\text{gt}} \cup D_{\text{inf}}$; confidence level δ
Ensure: Candidate θ_c or NSF
1: $D_V \leftarrow [TODO : \text{VerificationSplit}](D_{\text{gt}})$
2: $D_{\text{CS}} \leftarrow (D_{\text{gt}} \setminus D_V) \cup D_{\text{inf}}$
3: $\theta_c \leftarrow$ [TODO:] ▷ Candidate Selection
4: **return** $\text{Verification}(\theta_c, D_V, \delta)$

Algorithm 3 implements this modification. The verification set D_V is constructed exclusively from ground-truth samples, while the candidate selection set D_{CS} includes both the remaining ground-truth samples and all samples with inferred attributes. [TODO: discuss the candidate selection step.] As in Algorithm 2, the standard verification procedure is then applied to determine whether the candidate policy can be safely returned.

We now show that this modification results in an algorithm that preserves (1). As in Section 3.1, we instantiate the verification step using Student's t-test (Property 1) to construct HCUBs.

Theorem 3 (Verification with Inferred Attributes in Candidate Set)**.** *Let $D = D_{gt} \cup D_{inf}$. Suppose Assumption 1 and 3 hold. If Algorithm 3 is applied with (i) a random partition of D_{gt} into D_V and $D_{gt} \setminus D_V$, and (ii) $D_{CS} = (D_{gt} \setminus D_V) \cup D_{inf}$, then the probabilistic fairness guarantee in (1) holds.*

3.3 Inferred Attributes in Candidate and Verification Set

We next consider the regime in which samples with inferred sensitive attributes also appear in the verification set. In contrast to Section 3.2, the verification step must now reason directly about uncertainty in group membership induced by attribute inference. As implied by Lemma 1, naively applying the standard verification procedure in this regime can invalidate the assumptions underlying HCUBs, and therefore break the probabilistic fairness guarantee in (1). In this section, we develop a modified verification procedure that explicitly accounts for attribute inference error and restores valid high-probability fairness coverage under Assumption 1.

Consider a fixed candidate policy θ_c returned by a candidate selection method. Our goal is to construct a valid HCUB for the fairness violation $g(\theta_c)$ when sensitive attributes in the verification set may be inferred rather than observed.

We treat attribute inference error as inducing a bounded distortion between fairness quantities computed using the latent attributes S and those computed using the inferred attributes $\widehat{S}$. We first construct a high-confidence bound using the proxy labels $\widehat{S}$, and then inflate this bound by a deterministic bias term that accounts for worst-case misclassification consistent with Assumption 1. This yields a conservative but valid HCUB for the true fairness violation.

We present the construction for reward disparity in the main text and defer the analogous treatment of demographic parity to Appendix [TODO:].

Define the reward-disparity verification target

$$g_{\mathrm{RD}}(\theta_c) \coloneqq |r_{s_1}(\theta_c) - r_{s_2}(\theta_c)| - \epsilon. \tag{21}$$

Let

$$\Delta_{\mathrm{RD}}(\theta_c) \coloneqq r_{s_1}(\theta_c) - r_{s_2}(\theta_c)$$

denote the latent signed group reward difference, where group membership is defined with respect to the true attribute S. When verification rows contain inferred attributes, we instead compute a proxy signed difference

$$\widetilde{\Delta}_{\mathrm{RD}}(\theta_c)$$

using the inferred labels $\widehat{S}$.

The verification strategy proceeds in two steps:

1. Construct a high-confidence upper bound on $|\widetilde{\Delta}_{\mathrm{RD}}(\theta_c)|$ using the standard verification procedure applied to proxy-labeled data.

2. Inflate this bound by a deterministic term that upper bounds the discrepancy between the latent and proxy signed differences.

The resulting bound is conservative by construction; we show below that, under suitable assumptions, it provides a valid high-confidence upper bound on the true fairness violation.

Verification statistics. We denote by $q_{\theta_c}(X, A, R)$ the per-sample reward verification statistic used to evaluate reward disparity for a fixed candidate policy θ_c (e.g., an importance sampling or doubly robust contribution).

Assumption 5 (Bounded verification signal)**.** *There exists a constant $q_{\max} > 0$ such that*

$$|q_{\theta_c}(X, A, R)| \leq q_{\max} \quad \textit{almost surely}.$$

Assumption 7 ensures that the influence of any single verification sample is uniformly bounded. This condition is mild and standard in finite-sample verification analyses. In our setting, it holds naturally because rewards are bounded and common off-policy estimators either inherit this boundedness or employ weight clipping to prevent extreme values.

Intuitively, this assumption guarantees that occasional attribute misclassification cannot cause arbitrarily large distortions in estimated group statistics. Instead, the worst-case effect of mislabeling a sample is limited by $q_{\max}$, which enables deterministic control of the bias introduced by inferred attributes.

For a verification dataset of size m, we write

$$q_i \coloneqq q_{\theta_c}(X_i, A_i, R_i), \quad i = 1, \ldots, m,$$

for the observed verification samples.

Define the latent and proxy signed differences as

$$\Delta_{\mathrm{RD}}(\theta_c) = r_{s_1}(\theta_c) - r_{s_2}(\theta_c), \qquad \widetilde{\Delta}_{\mathrm{RD}}(\theta_c) = \widetilde{r}_{s_1}(\theta_c) - \widetilde{r}_{s_2}(\theta_c),$$

where $\widetilde{r}_s(\theta_c)$ denotes the group-conditional mean computed using inferred labels $\widehat{S}$.

Misclassification-induced bias. Attribute inference error induces a discrepancy between the latent and proxy disparities. Under bounded per-group misclassification error, this discrepancy can be bounded deterministically.

Assumption 6 (Bounded misclassification error)**.** *For each group $s \in \{s_1, s_2\}$,*

$$\Pr(\widehat{S} \neq S \mid S = s) \leq \varepsilon_s.$$

Assumption 7 (Bounded verification signal)**.** *The verification signal satisfies $|q_{\theta_c}(X, A, R)| \leq q_{\max}$ almost surely.*

Assumption 8 (Positive group mass)**.** *There exists $p_{\min} > 0$ such that $\Pr(S = s) \geq p_{\min}$ for all s.*

Under Assumptions 6–8, there exists a deterministic constant B_{mis} depending only on $\varepsilon_{s_1}, \varepsilon_{s_2}, q_{\max}$, and $p_{\min}$ such that

$$\left|\Delta_{\mathrm{RD}}(\theta_c) - \widetilde{\Delta}_{\mathrm{RD}}(\theta_c)\right| \leq B_{\mathrm{mis}}. \tag{22}$$

Misclassification-robust HCUB. Let $U_{\mathrm{proxy}}(\theta_c)$ be a valid $(1-\delta)$ high-confidence upper bound on $|\widetilde{\Delta}_{\mathrm{RD}}(\theta_c)|$ computed using the standard verification procedure with proxy labels. Define

$$U_{\mathrm{RD}}^{\mathrm{mis}}(\theta_c) \coloneqq U_{\mathrm{proxy}}(\theta_c) + B_{\mathrm{mis}} - \epsilon. \tag{23}$$

The theorem below shows that this inflated bound is a valid HCUB for the true reward-disparity violation.

Theorem 4 (Validity of misclassification-robust HCUB)**.** *Fix a candidate policy θ_c. Assume Assumptions 1 and 6–8. If $U_{\mathrm{proxy}}(\theta_c)$ is a valid $(1-\delta)$ HCUB for $|\widetilde{\Delta}_{\mathrm{RD}}(\theta_c)|$, then*

$$\Pr\big(g_{\mathrm{RD}}(\theta_c) \leq U_{\mathrm{RD}}^{\mathrm{mis}}(\theta_c)\big) \geq 1 - \delta.$$

Equivalently, $U_{\mathrm{RD}}^{\mathrm{mis}}(\theta_c)$ is a valid one-sided $(1-\delta)$ HCUB for the reward-disparity verification target under inferred attributes in the verification set.

4 Empirical Analysis

Baselines.

- **Lagrangian / fairness-constrained offline bandit (?).**
- **RobinHood (Seldonian constrained bandit (?)).** Let $a \in [L_a, U_a]$ and $b \in [L_b, U_b]$. Then to calculate the upper and lower bounds of $|a - b|$, RobinHood does the following:

$$a - b = a + (-b) \in [L_a, U_a] + [-U_b, -L_b] \tag{24}$$
$$= [\underbrace{L_a - U_b}_{L}, \underbrace{U_a - L_b}_{U}] \tag{25}$$

  Let $E = a - b$. Then $E \in [L, U]$.

$$|E| \in \begin{cases} [\min(0, |L|, |U|), \max(|L|, |U|)] & 0 \in [L, U] \\ [\min(|L|, |U|), \max(|L|, |U|)] & \text{otherwise.} \end{cases} \tag{26}$$

  The full candidate selection method for RobinHood is in Algorithm 4.
- Unconstrained offline policy optimization. Same policy class and off-policy estimator (e.g., DR) as our method, but without any fairness constraint. For the reference point for the fairness–utility tradeoff.
- Pessimistic offline policy optimization (**?**). Only skimmed the paper—just wanted to include a strong non-fairness-aware baseline.
- **Naive fairness unaware baseline**
- **Penalty-based fairness baseline.** Optimize

$$\max_{\pi} \ \widehat{V}(\pi) - \lambda \widehat{F}(\pi),$$

  sweeping λ to trace out a fairness–utility tradeoff curve, maybe.

Algorithm 4 RobinHood Candidate Selection Method

Require: Dataset D_{CS}; confidence level δ
Ensure: Candidate θ_c

```
Define objective f(θ):
    compute r̂(θ; D_CS)
    if U(θ) ≤ −ξ then f(θ) ← r̂(θ; D_CS)          ▷ obtain (inflated) constraint upper bound U
    else f(θ) ← r_min − max(0, U(θ))
Initialize CMA-ES parameters (m_0, σ_0, C_0 = I), population size λ, elites μ, weights w_{1:μ}
for t = 0, 1, 2, ... until termination do
    for j = 1 to λ do
        Sample z_j ~ N(0, I)
        Set θ_j ← m_t + σ_t A_t z_j                ▷ A_t A_t^⊤ = C_t
        Evaluate s_j ← f(θ_j)
    end for
    Sort {(θ_j, s_j)}_{j=1}^λ so s_1 ≥ ··· ≥ s_λ
    Select elites θ_{1:μ}
    Update mean m_{t+1} ← Σ_{i=1}^μ w_i θ_i
    Update evolution paths (p_σ, p_c)              ▷ standard CMA-ES
    Update covariance C_{t+1} using rank-1 and rank-μ updates   ▷ standard CMA-ES
    Update step size σ_{t+1}                       ▷ standard CMA-ES
end for
θ_c ← arg max_{θ∈{θ_j evaluated}} f(θ)             ▷ or use final mean m_T
return θ_c
```

5 Additional Related Work

- Methods for maintaining finite-sampel guarantees (Seldonian methods)
- Failure modes in seldonian? eg giguere work
- work in inferred attributes (methods to infer). methods to maintain fairness given proxy data

A Proofs for Section 3

Theorem 1 (Reduction to Verification Validity). *Consider Algorithm 1 with D_{gt} as input. Let θ_c be the candidate solution produced using D_{CS}, and let $U(\hat{g})$ be the HCUB of $g(\theta_c)$ computed on D_V. If $\Pr\big(g(\theta_c) > U(\hat{g})\big) \leq \delta$, then Algorithm 1 satisfies the fairness guarantee in (1).*

Proof. We bound the failure probability $\Pr\big(g(\mathrm{alg}(D)) > 0\big)$, where alg is Algorithm 1. Since by definition $g(\mathrm{NSF}) = 0$, the event $\{g(\mathrm{alg}(D)) > 0\}$ can occur only if the algorithm returns a candidate θ_c and that candidate violates the constraint. Acceptance of θ_c implies $U(\hat{g}) \leq 0$, so

$$\Pr\big(g(\mathrm{alg}(D)) > 0\big) \;\leq\; \Pr\big(g(\theta_c) > 0 \text{ and } U(\hat{g}) \leq 0\big).$$

But $\{g(\theta_c) > 0 \text{ and } U(\hat{g}) \leq 0\} \subseteq \{g(\theta_c) > U(\hat{g})\}$, hence

$$\Pr\big(g(\theta_c) > 0 \text{ and } U(\hat{g}) \leq 0\big) \;\leq\; \Pr\big(g(\theta_c) > U(\hat{g})\big) \;\leq\; \delta.$$

Therefore, $\Pr(g(\mathrm{alg}(D)) \leq 0) \geq 1 - \delta$. □

Theorem 2. Let g be defined as in Definition 1. Under Assumptions 3 and 4, $\Pr(g(\mathrm{alg}(D)) \leq 0) \geq 1 - \delta$, where alg denotes Algorithm 2.

Proof. **[1]** By Theorem 1, it suffices to show that the HCUB used in the verification step is statistically valid for the candidate solution θ_c, such that $\Pr\big(g(\theta_c) > U(\hat{g})\big) \leq \delta$. **[2]** Under Assumption 4 and the i.i.d. characteristic of the dataset D_{gt}, the candidate solution θ_c (which depends on D_{CS}) is statistically independent of the verification data D_V. **[3]** Because of this independence, and since the difference and absolute value operations in Definition 1 are applied to unbiased group-conditional expectations, the verification estimates $\hat{g}$ computed on D_V are conditionally i.i.d. and yield a valid estimate of $g(\theta_c)$. **[4]** By Assumption 3, the estimator $\hat{g}$ satisfies the method-specific regularity conditions required by the chosen HCUB $U(\cdot)$. **[5]** Therefore, the corresponding HCUB construction is statistically valid when applied to $\hat{g}$, implying $\Pr\big(g(\theta_c) > U(\hat{g})\big) \leq \delta$. Applying Theorem 1 yields the probabilistic fairness guarantee in (1). □

Lemma 1 (Failure of Naive Verification with Inferred Attributes). *Let $D = D_{gt} \cup D_{inf}$, where $D_{inf} \neq \emptyset$. If Algorithm 2 is applied under Assumption 4, then (1) need not hold.*

Proof. We construct counterexamples showing that the HCUB validity condition $\Pr\big(g(\theta_c) > U(\hat{g})\big) \leq \delta$ can fail when the verification set contains samples with inferred attributes. We present a single construction that applies to both fairness notions in Definition 1.

[1] Fix a candidate policy θ_c. Suppose the verification set D_V produced by Algorithm 2 under Assumption 4 contains n_{gt} samples from D_{gt} and n_{inf} samples from D_{inf}, with $n_{\mathrm{inf}} > n_{\mathrm{gt}}$. Let Z denote the per-sample quantity used in the verification step to estimate the constraint function $g(\theta_c)$. The estimator is the sample mean $\hat{g} = \frac{1}{|D_V|}\sum_{i=1}^{|D_V|} Z_i$. **[2]** Suppose that under the ground-truth data-generating process, $Z = 1$ almost surely, whereas for samples with inferred attributes, $Z = -1$ almost surely. Then $\hat{g} = \frac{1}{n_{\mathrm{gt}}+n_{\mathrm{inf}}}\Big(\sum_{i=1}^{n_{\mathrm{gt}}} 1 + \sum_{j=1}^{n_{\mathrm{inf}}}(-1)\Big) = \frac{n_{\mathrm{gt}}-n_{\mathrm{inf}}}{n_{\mathrm{gt}}+n_{\mathrm{inf}}} < 0$.

[3] For standard mean-based HCUBs, the confidence radius decreases to 0 as $|D_V|$ grows, while $\hat{g} < 0$ remains fixed. Hence for sufficiently large $n_{\mathrm{gt}} + n_{\mathrm{inf}}$, the upper bound satisfies $U(\hat{g}) < 0$. **[4]** The verification step accepts θ_c whenever $U(\hat{g}) \leq 0$, so the algorithm returns θ_c. However, under

the ground-truth distribution, $g(\theta_c) = \mathbb{E}[Z] = 1 > 0$, so the fairness constraint is violated. Hence $\Pr\big(g(\theta_c) > U(\hat{g})\big) = 1$, violating the HCUB validity requirement for any $\delta < 1$. Therefore, (1) need not hold.

For reward disparity, Z corresponds to the per-sample contribution to the estimated difference $r_{s_1}(\theta_c) - r_{s_2}(\theta_c)$; for demographic parity, Z corresponds to the per-sample contribution to the estimated difference $p_{s_1}(\theta_c) - p_{s_2}(\theta_c)$. If inferred attributes systematically swap group identities, these estimated group differences reverse sign, matching the construction above and producing the same verification failure. Thus the probabilistic fairness guarantee can fail for both fairness notions in Definition 1. □

Theorem 3 (Verification with Inferred Attributes in Candidate Set). *Let $D = D_{gt} \cup D_{inf}$. Suppose Assumptions 1 and 3 hold. If Algorithm 3 is applied with (i) a random partition of D_{gt} into D_V and $D_{gt} \setminus D_V$, and (ii) $D_{CS} = (D_{gt} \setminus D_V) \cup D_{inf}$, then the probabilistic fairness guarantee in (1) hold.*

Proof. **[1]** By Theorem 1, it suffices to show that the high-confidence upper bound used in the verification step is statistically valid for the candidate solution θ_c, i.e., $\Pr\big(g(\theta_c) > U(\hat{g})\big) \leq \delta$. **[2]** Under condition (i), D_{gt} is partitioned uniformly at random into D_V and $D_{\text{gt}} \setminus D_V$, independently of the sample values. Since D_{gt} is i.i.d., D_V is statistically independent of $D_{\text{gt}} \setminus D_V$. By Assumption 1, conditional on the fixed predictor h, the inferred-attribute dataset D_{inf} is independent of D_{gt}, and hence independent of D_V. Therefore, $D_V \perp \big((D_{\text{gt}} \setminus D_V) \cup D_{\text{inf}}\big) = D_{\text{CS}}$.

[3] By condition (ii), the candidate solution θ_c is computed using only $D_{\text{CS}} = (D_{\text{gt}} \setminus D_V) \cup D_{\text{inf}}$. Hence θ_c is a measurable function of D_{CS}, which implies $D_V \perp \theta_c$. Thus, the candidate solution is statistically independent of the verification data. **[4]** Because $D_V \subseteq D_{\text{gt}}$, the verification step uses only ground-truth sensitive attributes. Consequently, for a fixed candidate θ_c, the verification statistics computed on D_V are exactly those of the standard ground-truth verifier used in Theorem 2. In particular, they form conditionally i.i.d. unbiased estimates of the population fairness quantity underlying $g(\theta_c)$.

[5] By Assumption 3, the high-confidence upper bound construction used by the verifier is statistically valid for such ground-truth verification statistics. Therefore, the same HCUB argument as in Theorem 2 applies, yielding $\Pr\big(g(\theta_c) > U(\hat{g})\big) \leq \delta$. **[6]** Applying Theorem 1 gives $\Pr\big(g(\text{alg}(D)) \leq 0\big) \geq 1 - \delta$, so Algorithm 3 satisfies (1). □

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Supplementary Materials

The following content was not necessarily subject to peer review.

B Verification Step Using Hoeffding's Inequality

Consider a vector of m independent and identically distributed (i.i.d.) samples $(z_i)_{i=1}^m$ of a random variable Z; let the sample mean be $\bar{Z} = \frac{1}{m}\sum_{i=1}^m Z_i$, the sample standard deviation with Bessel's correction be $\sigma(Z_1, ..., Z_m)=\sqrt{\frac{1}{m-1}\sum_{i=1}^m (Z_i - \bar{Z})^2}$, and $\delta \in (0, 1)$ be a confidence level.

Property 2 (Hoeffding)**.** *If* $\Pr(Z \in [a, b]) = 1$*, then* $\Pr\left(\mathbb{E}[Z] \geq \bar{Z} - (b-a)\sqrt{\frac{\ln(1/\delta)}{2m}}\right) \geq 1 - \delta$.

Proof. See the work of Hoeffding (1963). □

Property 2 can be used to obtain a HCUB for the mean of Z:

$$U_{\texttt{Hoeff}}(Z_1, \ldots, Z_m) \coloneqq \bar{Z} + (b-a)\sqrt{\frac{\ln(1/\delta)}{2m}}. \tag{27}$$

Let $\hat{g}$ be a vector of i.i.d. and unbiased estimates of $g(\theta_c)$. These estimates can be given to $U_{\texttt{Hoeff}}$ to derive a HCUB on $g(\theta)$:

$$\Pr(g(\theta_c) \leq U_{\texttt{Hoeff}}(\hat{g})) \geq 1 - \delta. \tag{28}$$

Note that the validity of different methods for computing confidence intervals depends on specific assumptions, which must be satisfied for the intervals to be valid and for our theoretical guarantees to hold. For example, confidence intervals derived using Hoeffding's inequality require $\hat{g}(\theta_c)$ to be bounded.

[Define the verification step rule –Blossom]